

Dynatron

A bubble-and-jet modulator: flange, phase or a resonant filter, swept by an LFO, by hand or by your playing.



Overview

Dynatron is three modulation effects sharing one set of controls. FLANGE is a wide-sweep studio flanger with regeneration for the jet sound. PHASE is a six-stage phaser. FILTER is a resonant auto-wah / envelope filter with low-pass, band-pass and high-pass outputs.

What moves the sweep is up to you. LFO gives the usual automatic sweep. HOLD freezes it so you can place it by hand with CENTRE. ENV is the old two-tape-machine trick: your playing drives the sweep, so the harder you play, the further it goes.

A slow drift can make the rate, depth and centre wander, like a hand on the tape or an old clock.

Reach for it for

- Jet flanging and bubbling phase on guitars, synths and drum loops
- Touch-sensitive envelope filter and auto-wah on bass, funk guitar and keys
- Static comb or filter colour placed by hand (HOLD)
- Tempo-synced phasing
- Hollow, nasal tones with negative feedback

Controls

Mode, sweep source and filter type

FLANGE	A wide-sweep flanger. The delay sits between about 0.15 and 14 ms (set by CENTRE) and sweeps up to two octaves of delay either side. The loop is softened above about 9 kHz.
PHASE	A six-stage all-pass phaser, centred between 100 Hz and 2.4 kHz and sweeping up to three octaves either side.
FILTER	A resonant state-variable filter, centred between 150 Hz and 4 kHz and sweeping up to about three octaves either side. With RESONANCE up it squelches and nears self-oscillation.
LFO / HOLD / ENV	The sweep source. LFO: the usual automatic sweep. HOLD: the sweep stops and CENTRE places it by hand. ENV: your playing drives the sweep - louder notes push it further (fast attack, about 180 ms to settle back).
LP / BP / HP	FILTER mode only: low-pass, band-pass or high-pass. The keys are dimmed in the other modes.
DELAY / CUTOFF display	The live delay time (FLANGE, in ms) or the cutoff of the sweep (PHASE and FILTER, in Hz).

Sweep

These knobs work in all three modes.

CENTRE	Where the sweep sits at rest, 0 - 100%. In FLANGE, low settings give short delays and high, metallic combs; high settings lower, more chorus-like tones. In PHASE and FILTER it sets the centre frequency.
DEPTH	How far the sweep travels either side of CENTRE, 0 - 100%.

FEEDBACK / RESONANCE	FLANGE and PHASE: feedback, from -95% to +95%. Positive is thick, ringing and jet-like; negative is hollow and nasal. The level is compensated so strong settings don't jump out. In FILTER the knob becomes RESONANCE (the Q, shown as 0 - 95% either way round): turn it up for a sharper, squelchier peak.
SPREAD	Offsets the right side's LFO from the left, 0 - 180 degrees, for stereo width. It only applies with the LFO as the source.
SENS	ENV only: how hard you have to play to push the sweep. Higher settings react to quieter playing. It is greyed out with the other sources.

LFO

SIN / TRI / SQR / RND	The LFO shape: sine, triangle, square or random.
FREE / SYNC / FLOW	How the LFO rate is set: FREE in Hz, SYNC locked to the project tempo and timeline, FLOW tempo-synced but running continuously through loops.
RATE	The LFO speed: 0.02 - 10 Hz in FREE, or a note value from 4 bars to 1/32 (with dotted and triplet values) in SYNC. In FREE it reads 'stopped' while the sweep source is HOLD.

Master

MIX	Dry / effect balance, 0 - 100% (50% by default). Around 50% gives the deepest flange and phase notches; for the filter, higher settings put more of the filtered sound up front.
DRIFT	A slow random wander of the LFO rate, the depth and the centre, 0 - 100%. 50% is the standard amount, 100% twice that, 0 none; the wander also gets a little faster as you turn it up.
HPF	A high-pass filter on the effect only, from off (fully left) up to 500 Hz. Raise it to keep the low end clean and solid

	while the sweep happens higher up.
OUTPUT	Output level, -24 to +12 dB.
OUT meters	Output level, left and right, with a floating peak. Click a meter to clear the peak.

IN / OUT

TRIM	Input level before anything else, from -24 to +12 dB. A single click on the knob (without dragging) switches GUARD on or off; its LED at the knob's upper right is dark when GUARD is off, red when it is on, and glows brighter while it is catching peaks.
GUARD	On by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources.
SAT	Band saturation on the way out, 0 - 100%: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper (no latency), or a lookahead limiter that holds the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). The CLIP key flashes red while the clipper is working.

The header

PRESETS, arrows and display	PRESETS (or a click on the display) opens the list of factory presets; the ◀ / ▶ keys step through them. The display shows the preset number and name. A preset sets the mode, LFO, sweep and MIX controls; DRIFT, HPF, OUTPUT and the IN / OUT strip are left as they are. The last entry in the list resets everything to the defaults.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.

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Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Presets

Classic Bubble Flange	The default: FLANGE with a slow triangle sweep and moderate feedback.
Dyna Flange	FLANGE with a very slow, deep triangle sweep, strong feedback and 90 degrees of SPREAD.
Smooth Phase Sweep	PHASE with a gentle sine sweep and light feedback.
Square Throb Phase	PHASE with a slow triangle sweep, strong feedback and 120 degrees of SPREAD - a throbbing, wide phase.
Vocal Auto-Wah	FILTER (band-pass) with a slow sine sweep, low centre and 70% mix.
Random Bubble Filter	FILTER (band-pass) with a random LFO, full depth and high resonance - bubbling, unpredictable steps.
Stepped Jump Flange	FLANGE with a random LFO at 1.2 Hz and light feedback.
Synced Wide Phase	PHASE locked to quarter notes with the sides 180 degrees apart.
Envelope Squelch	FILTER (band-pass) driven by your playing (ENV), low centre, deep sweep and high resonance.
Hollow Negative Flange	FLANGE with a very slow sine sweep and negative feedback for a hollow, nasal tone.
Reset all to defaults	Every control back to its starting value.

Tips

- For the classic touch-sensitive swoosh, pick ENV and set SENS so only your harder notes push the sweep all the way.
- In HOLD, automate CENTRE for sweeps you control exactly, in time with the music.
- If the low end gets thin or wobbly, raise HPF so only the upper part of the sound is swept.
- Negative FEEDBACK in FLANGE gives a hollow tone that sits well on busy mixes; positive gives the jet.

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